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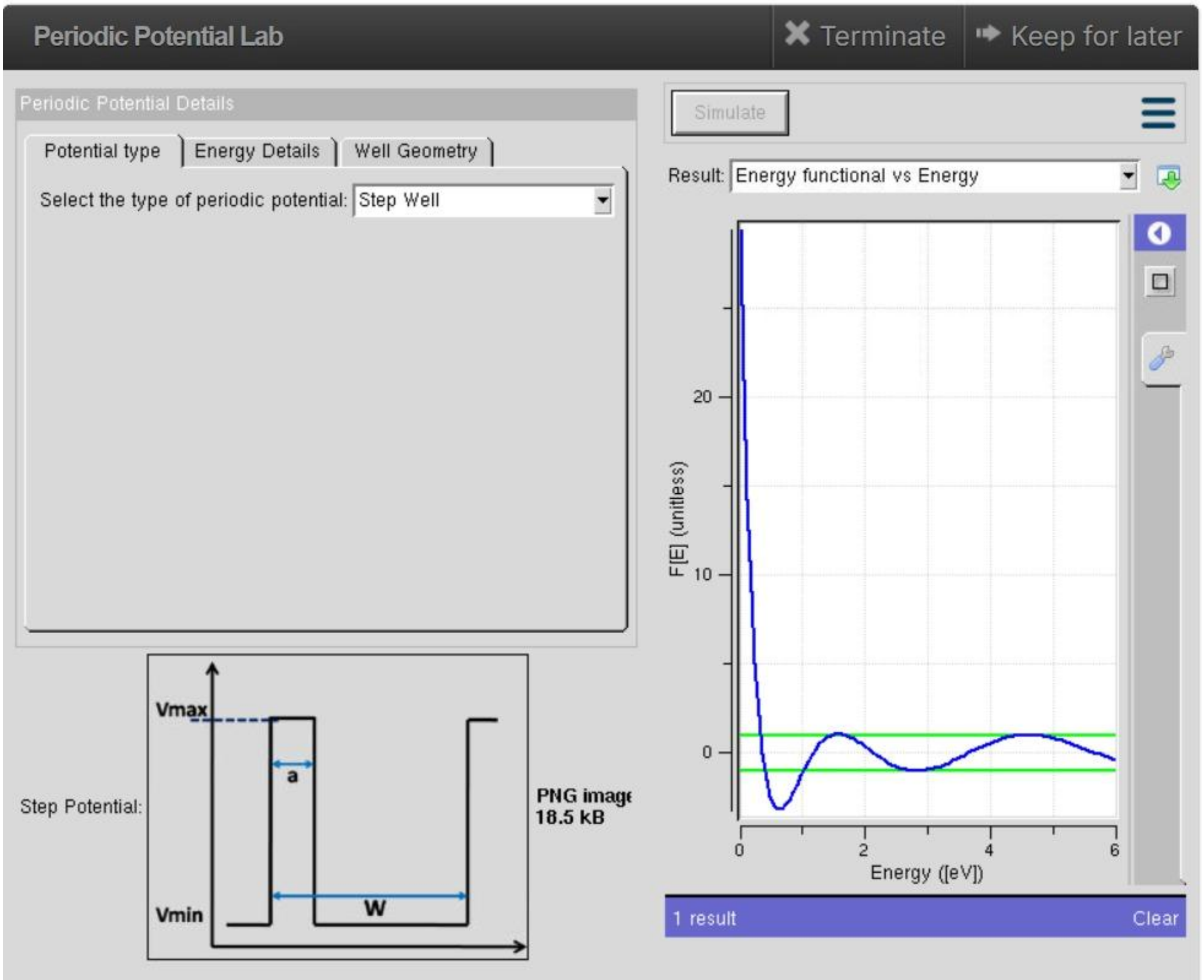
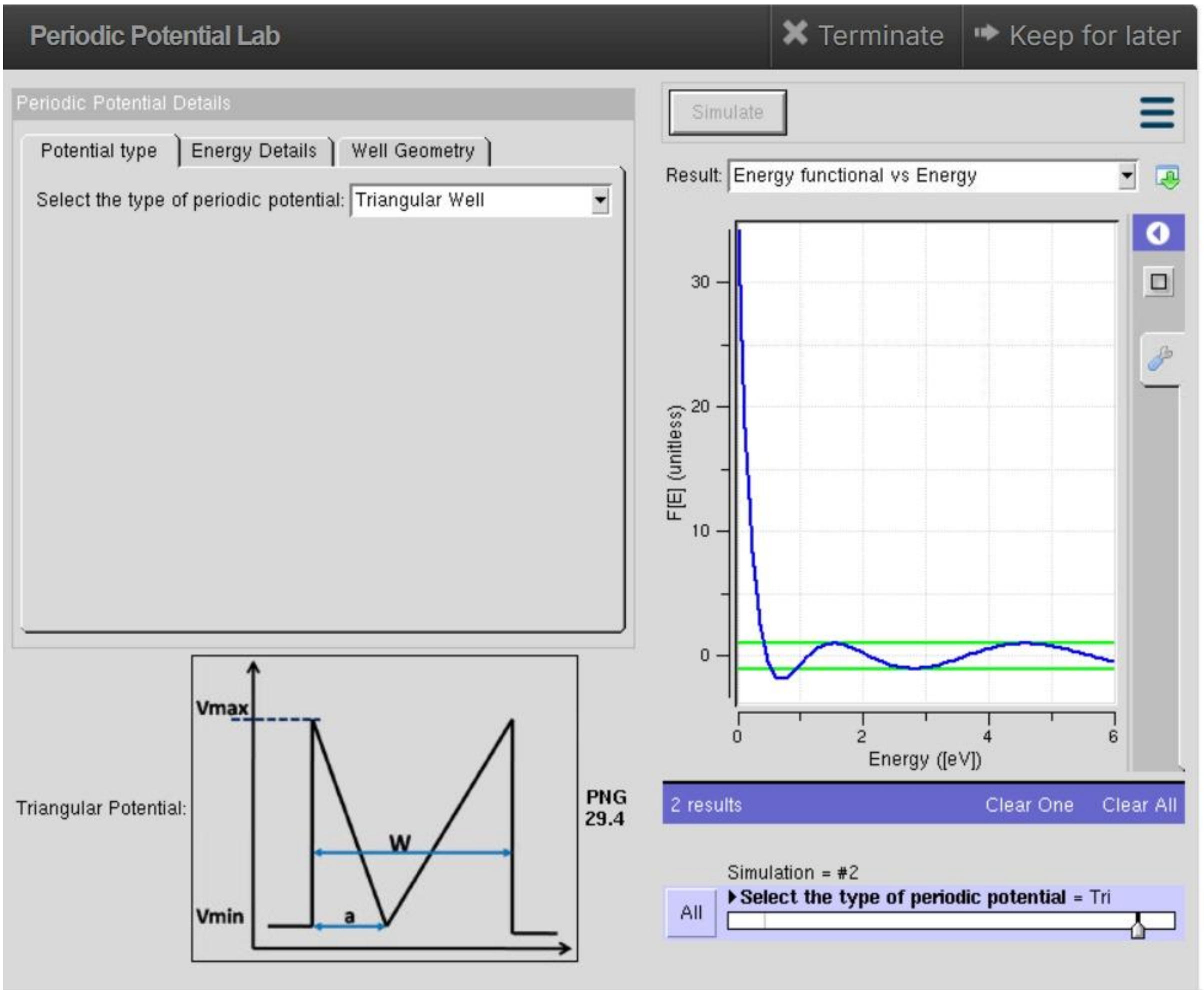
ECA3102 – NANO ELECTRONICS

EXPERIMENT. 4. Kronig Penney Model

OUTPUT:



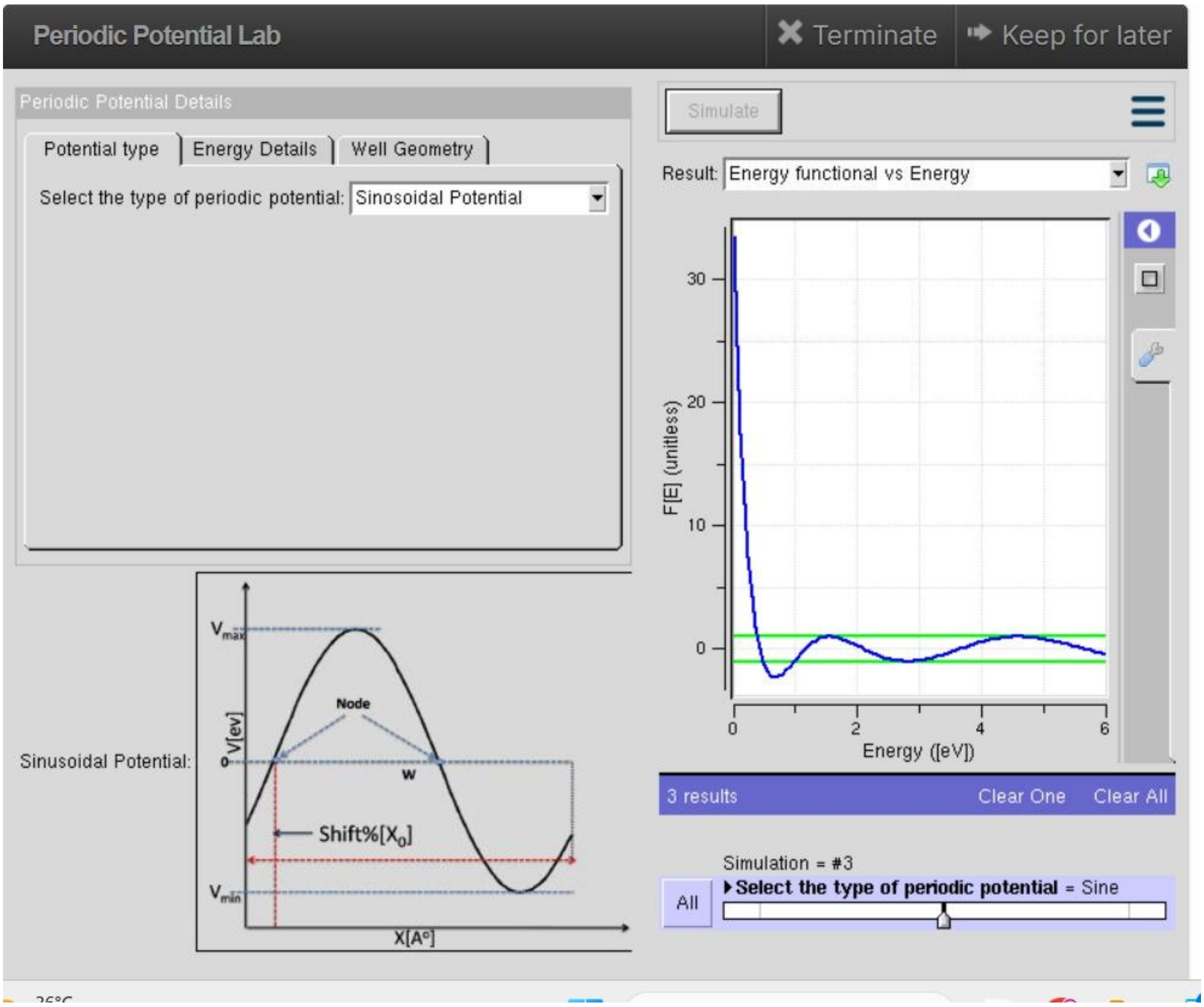
Case Study 1: Simulation of the output for Triangular well



Output Explanation

The simulation shows the **energy levels and wave functions** in a triangular quantum well. Due to the **triangular potential profile**, the energy levels are **non-uniformly spaced**. Electrons are confined near the lower potential region, resulting in **quantized energy states**. The wave functions are concentrated inside the well and decay outside the barrier.

Case Study 2: Simulation of the output for Sinusoidal Potential



OUTPUT:

The simulation demonstrates that a **sinusoidal potential** forms periodic quantum states and influences electron confinement. Such potential profiles are widely used in **superlattices, quantum devices, and nanostructures**.

CONCLUSION:

The simulations show that both the **Triangular Quantum Well** and **Sinusoidal Potential** produce **quantized energy levels** due to quantum confinement. The potential shape influences electron behavior, making these structures useful for **semiconductor and nanoelectronic applications**.